Lucas Mercier

PhD Candidate in Computer Vision applied to 3D Human pose estimation · Seeking Postdoctoral Position from late 2026

Laboratoire d'Innovation ouverte en technologie de la santé · ÉTS Montréal

GitHub LinkedIn Google scholar ResearchGate

Portfolio https://www.lucasmercier.me/

RESEARCH INTEREST

Computer vision, 3D Human pose estimation, Gait analysis, Machine learning, Image and video processing, Motion analysis, AI for healthcare

EDUCATION

École de technologie supérieure (ÉTS) - Montréal, Canada — Oct 2021 - Present

- Ph.D. in computer science, supervised by Carlos Vázquez, Neila Mezghani, Thierry Cresson
- Developed acquisition software to synchronize RGB and MoCap systems; built inpainting pipeline using GANs to remove reflective markers from images; benchmarked and improved SOTA monocular 3D pose estimation models for clinical gait parameters.
- Co-supervision of two 6 months internships (camera calibration tool & on edge pose estimation)

Institut supérieur d'électronique de Paris (ISEP) - Paris, France — 2016 - 2021

- Engineering school (equiv. M.Sc.) - Specialization: Data analysis and artificial intelligence (years 2 & 3)
- End of track research project: Myelin sheath segmentation using UNet for g-ratio estimation in neurological pathology analysis

Wrexham University - Wrexham, UK — Jan 2018 - May 2018

- Exchange semester - Software and mobile development

EXPERIENCE

Visiting PhD Student - **Institut de Biomécanique Humaine Georges Charpak** — Jan 2026 - Mar 2026

- Finetuned and benchmarked SOTA pose estimation model for gait parameter estimation on pathological populations
- Improved these 2D-to-3D lifting models by integrating visual features from RGB frames using multi-scale deformable offset sampling (manuscript in preparation)

Real-time Markerless Pose Estimation for Movement Coaching - **Flowtion (Co-founder) - Funded by Austria Wirtschaftsservice (55k€)** — Sept 2024 - Present

- Led technical architecture and development as the primary engineer on a two-person engineering team
- Built server-side pose estimation pipeline using WebRTC (aiortc) to stream client video for inference with pose models (HRNet, ViTPose etc.)

Pose Estimation Benchmarking Framework - **Developed for industrial partner (closed-source)** — 2023 - Present

- Unified plugin-oriented framework for training and evaluating SOTA 2D/3D pose estimation models. Reimplemented more than 15 models (HRNet, ViTPose, MotionBERT, MixSTE etc.). Designed for reproducibility and fair cross-model comparison.
- Used across multiple PhD projects and an industry collaboration for clinical markerless motion analysis.

R&D Computer Vision Intern - **Storelift** Feb 2021 - Aug 2021

- Implemented SOTA multi-object tracking algorithms (MOTNeuralSolver, FairMOT) for person tracking in autonomous retail environments.
- Developed a cross-camera object re-identification pipeline using TorchReID.
- Built a benchmarking framework to compare algorithm performance across metrics and camera configurations.

TEACHING EXPERIENCE

Teaching assistant - ETS - Montreal

- **GT411 - Digital Imaging** (Fall 2024, Winter 2025, Fall 2025 - 107 hours total)

GRANTS & AWARDS

Mobility Scholarship at Institut de Biomécanique Humaine Georges Charpak - Société de Biomécanique (International French-speaking Society of Biomechanics) 2025

Excellence grant for scientific communication activities - ÉTS Library 2024

SKILLS

Languages: Python, JS & TypeScript, SQL / Postgres, JAVA, C#
CV & ML: PyTorch, TensorFlow, OpenCV, scikit-learn
Tools: Git, Docker, Linux, SLURM & HPC (Compute Canada)

LAB IMPLICATION

Authored lab software development quality procedures (reproducibility standards, Git workflows, onboarding documentation) as part of ISO certification maintenance · **LIO-ÉTS** 2023-2025

PUBLICATIONS

JOURNALS

2026 Mercier L.,Cresson T. ,Gervais S. ,Mezghani N. ,Vázquez C. , Effect of markers in training dataset for markerless applications in biomechanics, Journal of Biomechanics [DOI]

CONFERENCES

2026 Mercier L.,Gajny L. ,Cresson T. ,Mezghani, N. ,Vázquez, C. , Investigating reflective marker impact in a multi-camera markerless motion capture scenario. , Congress of the French society of biomechanics · Montpellier, France 2026 [DOI]

2026 Mercier L.,Marois B. ,Fuentes A. ,Cagnin A. ,Cresson T. ,Vázquez, C. , Using knee kinesiography to train markerless pose estimation models for first-line kinematic assessment in knee osteoarthritis, OARSI World Congress on Osteoarthritis · Palm Beach, USA 2026 [DOI]

2025 Mercier L.,Cresson T. ,Mezghani N. ,Vázquez C. , Framework for the application of markerless motion capture to biomechanics, Congress of the French society of biomechanics · Marseille, France 2025 [DOI]

2025 Naaïm A. ,Rozaire J. ,Mercier L.,Begon M. ,Cherni Y. , Markerless motion capture accuracy in children with cerebral palsy and typically developing children, Congress of the French society of biomechanics · Marseille, France 2025 [DOI]

2025 Naaim A. ,Rozaire J. ,Mercier L.,Duprey S. ,Begon M. , Evaluation of 3D marker-less motion capture precision in upper limb children movement, International Society of Biomechanics · Stockholm, Sweden · 2025

2025 Mercier L.,Cresson T. ,Mezghani N. ,Vázquez C. , Étude de l'impact de la présence des marqueurs pour l'estimation de la pose humaine, 92e Congrès de l'Acfas · Montréal, Canada · 2025 [ORAL]